SE LAB -1

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Requirements Table

Problem Statement #20: Hospital Bed & ICU Allocation Dashboard

Actors: Triage Nurse, Hospital Administrator, Ambulance Dispatcher

Note: FR – 001 and NFR – 001 were provided beforehand.

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute a triage severity score from intake vitals and dynamically order the ER bed allocation waitlist by descending acuity.	High	Pass: A higher-acuity patient automatically moves ahead of a lower-acuity patient in the waitlist. Fail: A FIFO queue ignores a critical emergency.	Ensures the sickest patients are seen and beds are assigned first, reducing preventable deterioration.
FR-002	Functional	The system shall automatically allocate the next available ICU bed to the highest-priority waiting patient whose severity score exceeds the ICU admission threshold.	High	Pass: When an ICU bed status changes to Available, the system assigns it to the highest-scoring eligible waiting patient within 2 seconds and removes that patient from the waitlist. Fail: A bed remains unassigned for more than 2 seconds while an eligible patient is waiting.	Removes manual bed-matching delay, the main bottleneck in ICU transfer during surge conditions.
FR-003	Functional	The system shall allow a Triage Nurse to update the status of any ward or ICU bed to Occupied, Cleaning, or Available.	High	Pass: Selecting a new status for a bed updates its state in the database and the state is reflected on the dashboard. Fail: A bed status change made by a nurse is not saved or is overwritten silently.	Accurate bed state is the foundation all allocation and occupancy logic depends on.
FR-004	Functional	The system shall allow a Hospital Administrator to initiate an inter-facility ambulance transfer request for a patient, including the patient's current severity score and a check of destination-hospital bed availability before dispatch is confirmed.	Medium	Pass: A transfer request cannot be marked Confirmed until the destination facility's bed-availability check returns Available. Fail: A transfer is confirmed and dispatched to a facility with no free bed.	Prevents ambulances from being dispatched to facilities that cannot actually receive the patient.
FR-005	Functional	The system shall provide a Hospital Administrator with an aggregated occupancy view showing bed counts by status (Occupied, Cleaning, Available) per ward, filterable by ward and time range.	Medium	Pass: Selecting a ward and a date range returns occupancy counts that match the underlying bed-status log for that period. Fail: Filtered counts do not match the underlying log, or the filter has no effect on displayed data.	Gives administrators the capacity visibility needed for staffing and surge-planning decisions.

Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Bed status transitions (Occupied, Cleaning, Available) must propagate to all connected ER dashboard clients in under 1 second via WebSockets.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Stale bed status directly delays ICU allocation and can cost lives during surge events.
NFR-002	Security	The system shall enforce role-based access control so that only Hospital	High	Pass: A nurse account attempting an admin-only action (confirm transfer /	Protects patient-placement decisions from

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		Administrators can confirm ambulance transfers or override bed allocations, and shall log every bed-status change, allocation, and transfer action with a user ID and timestamp.		override allocation) is rejected with an authorization error, and every such action appears in the audit log with actor and timestamp. Fail: A nurse account can perform an admin-only action, or an action is missing from the log.	unauthorized changes and satisfies healthcare audit-trail requirements.

Use-Case Flow Specification

Problem Statement #20: Hospital Bed & ICU Allocation Dashboard

Use case name: Allocate ICU Bed

Primary Actor: Hospital Administrator

Preconditions

At least one patient is on the ER waitlist with a computed triage severity score (from UC-01).

The Triage Nurse has recorded intake vitals for every waitlisted patient.

The Hospital Administrator is authenticated on the dashboard.

Postconditions

The selected ICU bed's status changes from Available to Occupied.

The assigned patient is removed from the ER waitlist and linked to the ICU bed record.

An allocation event (patient ID, bed ID, administrator ID, timestamp) is written to the audit log.

Main Success Scenario

1. Hospital Administrator opens the ICU Allocation panel on the dashboard.
2. System includes UC-01 (Compute Triage Severity Score) to refresh the current severity score for every waitlisted patient.
3. System displays the waitlist ordered by descending severity score, alongside the list of ICU beds currently marked Available.
4. Administrator selects an Available ICU bed.
5. System identifies the highest-priority waitlisted patient who meets the ICU admission threshold for that bed.
6. System proposes the match (patient ↔ bed) and displays a confirmation prompt.
7. Administrator confirms the allocation.
8. System updates the bed status to Occupied and removes the patient from the waitlist.
9. System writes the allocation event to the audit log and broadcasts the updated bed status to all connected dashboard clients via WebSocket.
10. Use case ends successfully.

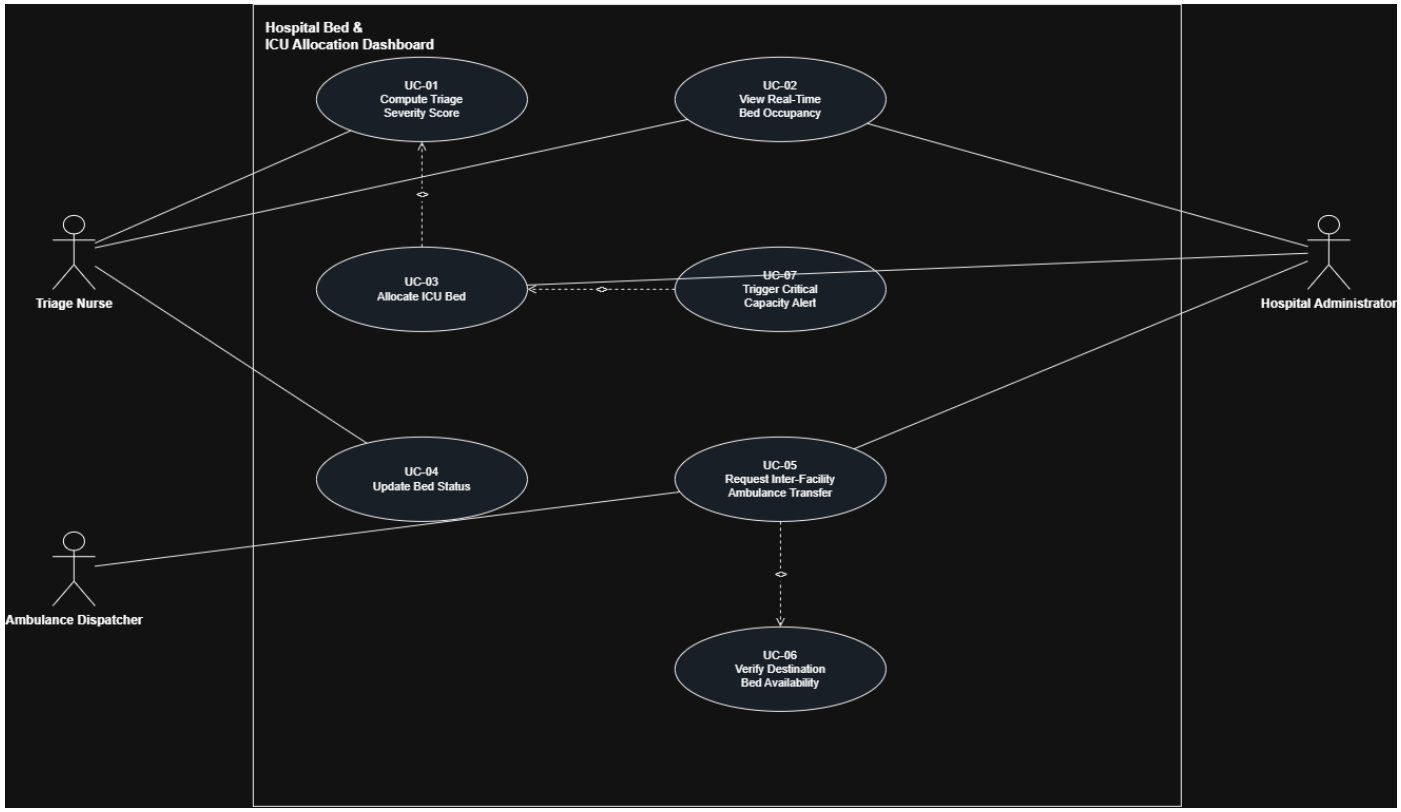
Alternate Flow

6a. No Eligible Patient for Selected Bed

6a1. System finds no waitlisted patient whose severity score meets the admission threshold for the selected bed type.

6a2. System displays a "No eligible patient" notice and leaves the bed status as Available.

6a3. Administrator may select a different bed or wait for the waitlist to update; the use case ends without an allocation.


Hospital Bed &
ICU Allocation Dashboard
UC-01
Compute Triage
Severity Score
UC-02
View Real-Time
Bed Occupancy
UC-03
Allocate ICU Bed
UC-07
Trigger Critical
Capacity Alert
UC-04
Update Bed Status
UC-05
Request Inter-Facility
Ambulance Transfer
UC-06
Verify Destination
Bed Availability
Triage Nurse
Hospital Administrator
Ambulance Dispatcher